



ASTER

Adaptive Smart Traffic Event Response

"Forecast the disruption before it disrupts."

Demo Video Script & Presentation Guide

Hackathon Submission · Event-Driven Congestion Track
City: Bengaluru, Karnataka · Dataset: 8,173 Real Events

■ Duration	5 min 30 sec target
■ Model	Gradient Boosting + LightGBM Cascade
■ Accuracy	93.1% (Post Leakage Fix — Real Numbers)
■ Run	streamlit run app/aster_app.py

A · PROJECT ANALYSIS

Project Summary

Field	Details
Project Name	ASTER — Adaptive Smart Traffic Event Response
Tagline	"ASTER turns incident reports into response plans — in under 2 seconds."
Problem	Bengaluru: 7M+ vehicle trips/day, 8,000+ incidents/year handled purely by officer experience — no prediction, no standard protocol
Target Users	BTP Operations Officers · Traffic Control Room (TCR) staff · City Traffic Management Authorities
Main Solution	ML classifier + decision engine → impact tier (Low/Med/High) + officers + barricades + diversion + SLA + checklist, in <2 seconds
Dataset	8,173 real events from ASTRAM system (Nov 2023 – Apr 2024)
Tech Stack	Python · Gradient Boosting (sklearn) · LightGBM Cascade · Google OR-Tools MILP · NetworkX routing · SHAP · Streamlit · Pydeck / Leaflet
Key Accuracy	GB: 93.1% accuracy · 91.5% F1-macro · AUC-ROC 0.97 (post leakage fix — real, honest numbers)
Setup Command	streamlit run app/aster_app.py

Key Features

Feature	What It Does
1. Dual-Model Prediction	GBM for impact tier + LightGBM cascading pipeline for resolution time, road closure probability, and Event Impact Score (EIS)
2. SHAP Explainability	Top 3 drivers shown for every prediction in plain language — not a black box
3. Tactical Map	Leaflet map with incident pin, barricade markers, and alternative diversion routes
4. MILP Manpower Optimizer	Google OR-Tools allocates officers across concurrent events in real time
5. Citizen Advisory Generator	One-click WhatsApp, Twitter/X, and VMS board message drafts
6. Pre-Event Planner	Forward-looking deployment timeline for scheduled events (matches, rallies, festivals)
7. Post-Event Learning	Model drift tracker, feedback logging, one-click retraining pipeline
8. Live Feed Simulator	Sidebar button simulates live ASTRAM alert pop-ups

Best Demo-Worthy Pages (in order)

#	Page	Why Show It
1	■ Overview	Thumbnail / First impression. KPI cards + 3D heatmap.
2	■ EDA → Spatial tab	Corridor Risk Calendar heatmap — best single chart in the app.
3	■ Predict & Respond	MAIN PAGE. Run Bus Breakdown + Water Logging presets. 2.5 min here.
4	■ Model Performance	Credibility moment. Show confusion matrix + honest leakage note.
5	■ Pre-Event Planner	Future scope story. Enter IPL match and generate T-minus timeline.

B · DEMO VIDEO FLOW (5 min 30 sec)

Timestamp Overview

Time	Section	What to Do
0:00 – 0:30	Hook + Problem Statement	Speaking only. No screen needed. Or show Overview frozen.
0:30 – 1:00	Overview — KPI Cards	Show 6 KPI cards + 3-tier color system. Mention 8,173 events.
1:00 – 1:30	Overview — Live Heatmap	Scroll to pydeck heatmap. Tilt/zoom to show 3D Bengaluru.
1:30 – 2:00	EDA → Spatial → Corridor Heatmap	Navigate to EDA, click Spatial tab, scroll to Corridor Risk Calendar.
2:00 – 4:25	Predict & Respond (CORE)	Bus Breakdown preset → Water Logging preset → Citizen Advisory → Live Alert.
4:25 – 5:00	Model Performance	Metrics + confusion matrix + leakage transparency note.
5:00 – 5:20	Pre-Event Planner	IPL match example. T-minus deployment timeline.
5:20 – 5:30	Closing Line	Deliver impact statement. End strong.

C · FULL NARRATION SCRIPT

Complete 5 min 30 sec Script

Speaking style: Confident, casual, like explaining to a smart friend. Never robotic. Pause after each key point. Keep energy up.

[0:00 – 0:30] Opening Hook

■ No screen needed — or show Overview page frozen. Speak directly to camera.

Every morning in Bengaluru, thousands of things go wrong on the road.

A BMTC bus breaks down on Mysore Road at 8 AM.

A procession blocks ORR during evening peak.

Water logging takes out a subway at Hebbal.

And every time — the traffic police response is the same.

An officer gets a call, uses their experience, and figures it out.

No prediction. No standard protocol. No data.

Just gut instinct, for over 8,000 events every single year.

We built ASTER to change that.

[0:30 – 1:00] Overview Page — KPI Cards + Three-Tier System

■ Open app → ■ Overview is the default page. Let it load. Point to the top metric cards.

This is ASTER's command dashboard.

8,173 real traffic events from Bengaluru's ASTRAM system.

13.5% of them caused High impact disruptions.

8.3% triggered full road closures.

ASTER works on a three-tier system.

Low — monitor only, 1 or 2 officers.

Medium — light barricading, advisory diversion.

High — full closure, 4 to 8 officers, under 5 minutes to deploy.

Simple to understand. Powerful in practice.

■ Scroll down slightly to show the three-tier color-coded cards on the right side.

[1:00 – 1:30] Overview — Live Density Heatmap

■ Scroll down on the Overview page to the 3D pydeck heatmap of Bengaluru.

*Here's every single event in our training data — plotted on a live map of Bengaluru.
You can see exactly where incidents cluster.
Mysore Road, ORR, Central Zone — these are Bengaluru's high-stress arteries.*

*You can right-click and drag to tilt the map, zoom into any area.
This isn't a mockup — this is real data, real coordinates.*

■ Zoom in once or tilt the map to show 3D perspective. Good visual pause — hold for 5 seconds.

[1:30 – 2:00] EDA → Spatial Tab — Corridor Risk Heatmap

■ Click ■ EDA & Insights in sidebar → click Spatial tab → scroll to Corridor Risk Calendar Heatmap.

Now let me show you one of my favorite insights from the data.

This is the Corridor Risk Calendar — it shows which corridors get hit with High-impact events, and at what time of day.

*Mysore Road — brutal during morning peak, hours 7 to 10.
ORR East — lights up in evening peak and on weekends.*

*This is the kind of pattern that normally lives in an officer's head.
ASTER puts it into a system anyone can act on.*

■ Pause on the heatmap for 5–8 seconds. This is a strong visual for judges.

CORE DEMO · Predict & Respond [2:00 – 4:25]

[2:00] Navigate to Predict & Respond

■ Click ■ Predict & Respond in sidebar.

*This is the core of ASTER — the Event Triage and Response Planner.
This is what a TCR officer would use in real operations.*

Scenario 1: Bus Breakdown on Mysore Road [2:00 – 3:15]

■ Click the ■ Bus Breakdown / Mysore Road 8 AM quick scenario button.

Let's run the classic scenario — a BMTC bus breaks down on Mysore Road at 8 AM morning peak.

■ While result loads, keep talking. Result appears in 2–3 seconds.

■ Point to LEFT card (Resolution Time):

Look at the resolution time forecast — LightGBM predicts around 45 to 60 minutes of expected delay.

*And on the right — Gradient Boosting calls it **High Impact** with over 90% confidence.*

Not a guess. Not gut feel. A model trained on 8,000 real Bengaluru events.

■ Scroll down slightly to the SHAP section — Top 3 Driving Factors.

*Here's what makes this different from any other traffic tool.
SHAP explainability — it tells you exactly why it made this call.
Top driver: this is on a named corridor, Mysore Road.
Second: morning peak hour — historically much higher risk.
Third: road closure flag.*

The officer doesn't just get an answer. They get the reasoning.

■ Scroll down to the Leaflet tactical map.

*And here — a tactical map.
The red dot is the incident. The amber dots are recommended barricade points.
The blue line shows the primary diversion route.
Officers know exactly where to go.*

■ Scroll down to the Operational Response Plan section.

Now the response plan.

Response Priority: P1 — Critical.

Officers allocated — optimized using Google OR-Tools to balance across all concurrent incidents in the city.

Deploy within 5 minutes. Mandatory diversion.

And the action checklist — step by step.

Coordinate tow truck. Notify TCR. Activate diversion.

Specific to a vehicle breakdown. Not a generic alert.

Scenario 2: Water Logging on ORR East [3:15 – 4:00]**■ Click the ■ Water Logging / ORR East 9 AM quick scenario button. Wait for result.**

Now let's switch to a completely different event type — water logging on ORR East at 9 AM.

Medium impact. Expected delay around 35 minutes.

Notice — the response plan is completely different.

BBMP drainage cell alert. Avoid the underpass. Alternate routing toward Marathahalli.

Different cause, completely different response.

ASTER doesn't serve the same generic plan for every event.

It knows what each event type needs.

Citizen Advisory Generator [4:00 – 4:15]**■ Click the Generate Public Advisory Draft button.**

One more thing — we generate citizen advisories automatically.

WhatsApp template, ready to send.

Twitter alert, ready to post.

VMS board message — 40 characters, exactly what they need.

One click — and the public is informed.

Live Feed Simulator [4:15 – 4:25]

■ In the SIDEBAR, click ■ Simulate Live Alert. A toast notification pops up.

And here — the live feed simulator.

In production, this would be a real ASTRAM webhook.

A new event fires. ASTER picks it up instantly and pushes the alert.

[4:25 – 5:00] Model Performance

■ Click ■ Model Performance in sidebar.

For the technical judges — here's our model performance.

*Gradient Boosting — 93.1% accuracy, 91.5% F1 macro.
AUC-ROC: 0.97.*

*And I want to be transparent about something important.
We had a target leakage issue in our first version.
We found it. We fixed it. The numbers went from 99.9% to 93%.*

We chose to keep the real numbers because honest engineering matters more than inflated metrics.

Any judge who looks carefully will respect that call.

■ Point to the confusion matrix image.

*Confusion matrix shows clean separation across all three tiers.
High impact events — the critical minority — still caught at 80% recall.
That's the number that saves the city.*

[5:00 – 5:20] Pre-Event Planner

■ Click ■ Pre-Event Planner in sidebar. Enter 'IPL Match - RCB vs CSK', set crowd to Large (15k-40k), click Generate.

One last thing before I wrap up.

*ASTER isn't just reactive. We can plan ahead.
An IPL match at Chinnaswamy. 30,000 people. Evening kickoff.
Click generate — and ASTER gives you a pre-deployment timeline.*

*T-minus 2 hours: set barricades.
T-minus 1 hour: deploy 12 officers and issue mandatory diversion advisory.
T-zero: full road closure.*

The intelligence goes from incident response to event planning.

[5:20 – 5:30] Closing

■ Stay on current screen or return to Overview. Speak directly to camera.

*ASTER is built entirely on real data. Every number is honest.
8,173 real events. Real accuracy. Real deployment logic.*

*It doesn't replace an officer's judgment.
It gives them the right information, in the right format, in under 2 seconds.*

Bengaluru gets smarter. Officers respond faster. Commuters move.

Thank you.

D · SHORT BACKUP SCRIPT (3-minute version)

3-Minute Condensed Script

Use this if setup runs long or you need to cut content.

[0:00 – 0:20] Hook

Bengaluru. 7 million trips a day. 8,000 traffic incidents a year. Officers decide everything by experience. ASTER changes that.

[0:20 – 0:45] Overview

8,173 real events. 3-tier system. KPI dashboard. Live heatmap of every event across the city.

[0:45 – 2:30] Predict & Respond

Click Bus Breakdown preset. Show impact card + SHAP + tactical map + response plan. Then click Water Logging. Show different cause, different plan.

[2:30 – 2:50] Model Performance

93.1% accuracy. We had leakage. We fixed it. Real numbers only.

[2:50 – 3:00] Close

Real data. Real model. Real plan. Under 2 seconds. That's ASTER.

E · BEST FEATURES TO HIGHLIGHT FOR JUDGES

Judge-Facing Selling Points

Feature	Why It Impresses Judges
Dual-model architecture	GBM + LightGBM cascade — two completely different ML approaches, each with a distinct purpose
SHAP explainability	Every prediction shows its top 3 drivers. Not a black box. Officers trust what they understand.
MILP manpower optimizer	Google OR-Tools operations research — depth that pure ML projects don't have
Leaflet tactical map	Visual output an actual officer could use on a tablet. Practical, not decorative.
Honest leakage disclosure	Found 99.9% was inflated, fixed it to 93%, and said so. Intellectual integrity = trust.
Cause-specific checklists	16 event types, each with a different action protocol. Operational realism.
Post-event learning loop	Model drift tracker + actual retraining pipeline. Shows production thinking.
Citizen advisory output	WhatsApp + Twitter + VMS. Multi-stakeholder design. Shows breadth.
Pre-event planner	Goes from reactive to proactive. Shows the full intelligence lifecycle.
Real ASTRAM dataset	Zero synthetic data. ASTRAM is a real Bengaluru Traffic Police operational system.

F · Suggested Closing Lines

Pick one or combine:

- Option 1:** *ASTER doesn't replace an officer's judgment. It makes that judgment faster, smarter, and consistent — every single time.*
- Option 2:** *This is not a research demo. This is a production-ready decision layer for any smart city's traffic operations.*
- Option 3:** *We built this on real data, with honest engineering, for officers who need results — not dashboards. That's ASTER.*
- Option 4:** *8,000 incidents. 2 seconds per response. That's the difference ASTER makes.*

Option 5: Bengaluru taught us what the data looks like. ASTER teaches the system what to do about it.

G · PRE-RECORDING CHECKLIST

Before You Hit Record

Stat us	Check Item
■	Run streamlit run app/aster_app.py from project root — confirm it opens on localhost:8501
■■	Overview page: all 6 KPI cards visible (Total Events, High Impact, Planned Events, Road Closures, Corridors Affected, Median Resolution)
■■	Overview heatmap: pydeck renders correctly — needs working internet for map tiles (use wifi, not mobile hotspot)
■■	EDA → Spatial → scroll to bottom — confirm Corridor Risk Calendar heatmap renders fully
■	Predict & Respond → click Bus Breakdown → confirm: resolution time card, impact tier card, SHAP, leaflet map, manpower table, action checklist all appear
■	Predict & Respond → click Water Logging → confirm a different action checklist appears
■■	Click Generate Public Advisory Draft — confirm WhatsApp, Twitter, VMS messages appear
■■	Sidebar → click Simulate Live Alert → confirm toast notification pops up
■	Model Performance — confirm confusion_matrix.png loads (check assets/ folder)
■■	Pre-Event Planner: enter IPL match, crowd = Large, click Generate — confirm T-minus timeline appears
■	BEST THUMBNAIL: Predict & Respond after Bus Breakdown — HIGH IMPACT card + tactical map visible together
■	BEST PAUSE POINT: SHAP section (Top 3 Driving Factors) — clear, visual, impressive for judges
■	Fresh machine? Run: pip install -r requirements.txt first
■■	Set browser zoom to 90% — all columns fit without horizontal scroll
■■	Record in a quiet room. The script has natural pauses — use them.

H · Page Order Summary

Ste p	Page	What to Do
1	■ Overview	Let load. Point to KPI cards. Mention 3-tier system. Scroll to heatmap, tilt it.
2	■ EDA & Insights	Click Spatial tab. Scroll to Corridor Risk Calendar. Pause here 5–8 sec.

Step	Page	What to Do
3	■ Predict & Respond	Bus Breakdown preset → SHAP + map + plan + checklist → Water Logging preset → Public Advisory → Live Alert (sidebar).
4	■ Model Performance	Metrics. Confusion matrix. Leakage note. Honesty moment.
5	■ Pre-Event Planner	IPL match entry. Generate. T-minus timeline. Future scope.
END	Any page	Deliver closing line. End with energy.

ASTER — Adaptive Smart Traffic Event Response · Built for Bengaluru · Demo Video Script prepared for Hackathon Submission 2024